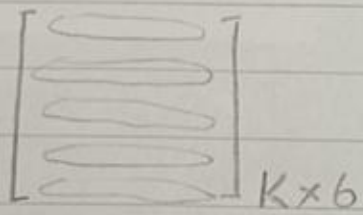
$\Theta = [\theta_1, \theta_2, \cdots, \theta_i, \cdots, \theta_I]$

θ_i表示学习者i的能力特征

$\eta = [\eta_1, \eta_2, \cdots, \eta_i, \cdots, \eta_I]$

η_i表示学习者i的努力特征

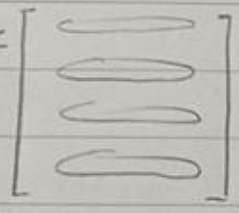
$\Psi =$ /psai/



$K \times 6$

知识特性矩阵，K表知识点数量

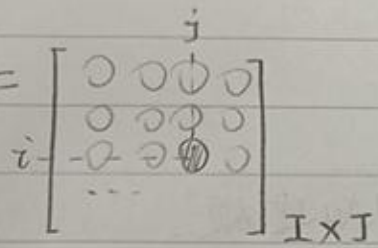
$\Omega =$



$J \times 3$

试题特性矩阵，J表试题数量

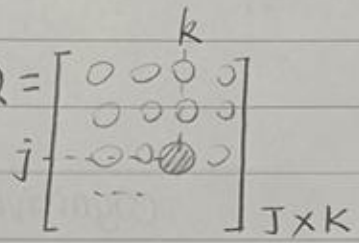
$\alpha =$



$I \times J$

学习者i对试题j的知识水平

$Q =$

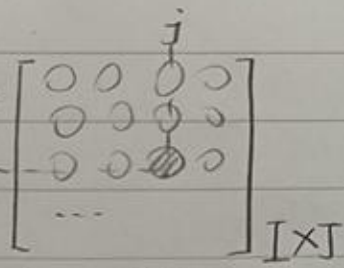


$J \times K$

j试题考查了知识点k为1，否则为0

试题与知识点的关联矩阵

$r =$



$I \times J$

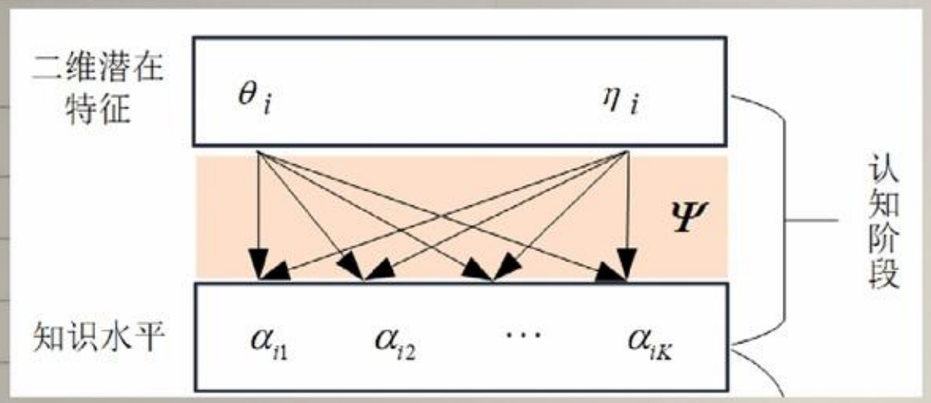
学习者i对试题j的作答结果

在Matlab代码中

"I"学习者数量：stu_num

"J"试题数量：item_num

"k"知识点数量：knw_num



认知阶段建模 @刘畅P21

$$\alpha_i = f(\theta_i, \Psi, \eta_i)$$

$\alpha_i = [\alpha_{i1}, \alpha_{i2}, \cdots \alpha_{iK}]$ $1\times K$; θ_i 1×1 1个数 ; Ψ $K\times6$; η_i 1×1

@刘畅P33 $\Psi_k = [w_{k1}, w_{k2}, d_{\theta k1}, d_{\eta k1}, d_{\theta k2}, d_{\eta k2}]$

某行 第k行

省略k, $\Psi = [\underbrace{w_1, w_2}_{\text{和为1}}, \underbrace{d_{\theta1}, d_{\eta1}, d_{\theta2}, d_{\eta2}}_{\text{在能力特征和努力特征上的难度参数,服从正态分布}}]$

cognitive_m1.m "认知建模"

```
function alpha=cognitive_m1(a2,d211,d212,d221,d222,theta,yita)
stu_num=length(theta);
knw_num=length(a2);
%%%%%%%%%%%%%%% 复制操作
a2_cop=repmat(a2,stu_num,1);%将w1的值复制了~~466~~42行给a2_cop
d211_cop=repmat(d211,stu_num,1);%theta第一阈值
d212_cop=repmat(d212,stu_num,1);%yita第一阈值
d221_cop=repmat(d221,stu_num,1);%theta第二阈值
d222_cop=repmat(d222,stu_num,1);%yita第二阈值
theta_cop=repmat(theta',1,knw_num);
yita_cop=repmat(yita',1,knw_num);
%%%%%%%%%%%%%%%%%%%%%%%%
lambda_theta=1./(1+(1-a2_cop).*exp(-1.7*(yita_cop-d222_cop)));
lambda_yita=1./(1+a2_cop.*exp(-1.7*(theta_cop-d221_cop)));
alpha_theta=a2_cop./(1+exp(-1.7*(theta_cop-d211_cop)));
alpha_yita=(1-a2_cop)./(1+exp(-1.7*(yita_cop-d212_cop)));
alpha=(lambda_theta.*alpha_theta+lambda_yita.*alpha_yita);
%alpha=(alpha-repmat(mean(alpha,1),stu_num,1))./(repmat(std(alpha,1,1),stu_num,1));
end
```

w_1 $d_{\theta1}$ $d_{\eta1}$ $d_{\theta2}$ $d_{\eta2}$ θ η

@刘畅P27 (3.13)

式(3.14)

式(3.12)

该函数的输入参数都是行向量。$w_1, d_{\theta1}, d_{\eta1}, d_{\theta2}, d_{\eta2}$ 是知识参数，故有 knw_num 列，简称K列；θ, η 是学习者参数，有 stu_num 列，简称S列

例：$\theta = [\theta_1, \theta_2, \cdots, \theta_{stu_num}]$，$w_1 = [v_1, v_2, \cdots, v_{knw_num}]$

repmat(A, r, c) 函数实现将矩阵A整体作为r行c列的矩阵的元素，返回1个新的矩阵。

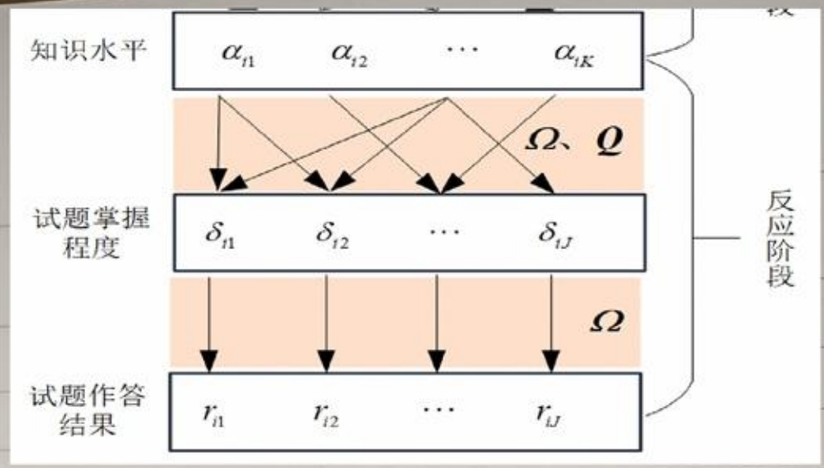
如：$a2_cop = \begin{bmatrix} v_1 & v_2 & \cdots & v_K \\ v_1 & v_2 & \cdots & v_K \\ \vdots & \vdots & & \\ v_1 & v_2 & \cdots & v_K \end{bmatrix}_{S\times K}$ $\quad$ $theta_cop = \begin{bmatrix} \theta_1 & \theta_1 & \cdots & \theta_1 \\ \theta_2 & \theta_2 & \cdots & \theta_2 \\ \vdots & \vdots & & \vdots \\ \theta_S & \theta_S & \cdots & \theta_S \end{bmatrix}_{S\times K}$

std(alpha, w, dim)：按dim求std(标准差)，w=1不归一化

故 $std(alpha, 1, 1) = [std_{11}, std_{12}, \cdots, std_{1K}]$

沿第1维计算每列的标准差

∴ alpha矩阵中每个元素* $= \dfrac{\text{该元素一列均值}}{\text{列标准差}}$ "$\dfrac{x-\mu}{\sigma}$"



response_m1.m "反应建模"

```
function [r,R]=response_m1(alpha,a1,s1,g1,T)
stu_num=size(alpha,1);
item_num=length(s1);
s1_cop=repmat(s1,stu_num,1);%
g1_cop=repmat(g1,stu_num,1);
alpha_cop=repmat(alpha,1,1,item_num);
alpha_cop=permute(alpha_cop,[1,3,2]);%permute重新排列维数
a1_cop=repmat(a1,1,1,stu_num);
a1_cop=permute(a1_cop,[3,2,1]);
temp_yita=(1-alpha_cop).*a1_cop;%temp_yita知识弱项参数
%temp_yita(temp_yita==0)=inf;
r1=(1-s1_cop).*(1-max(temp_yita,[],3));
r2=g1_cop.*(max(temp_yita,[],3));
r=r1.*T+r2.*T;
% r=1./(1+exp(-1.7*(alpha*(Q'.*a1)-d1_cop)));
r_rand=rand(size(r));%生成466*90的0-1之间均匀分布的随机数矩阵
R=r_rand<=r;
R=double(R);
R(T~=1)=-1;
end
```

T是stu_num行item_num列的矩阵，元素0表示作答，1表示作答了

试题数量

P32式(3.21)

式(3.23)的$(1-s_j)\delta_{ij}$

式(3.23)的$g_j(1-\delta_{ij})$

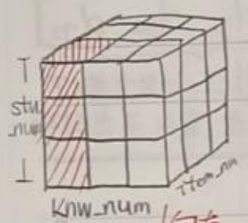
$(i\times j)$ r试题作答结果正确的概率矩阵

stu_num × item_num

因精度信息丢失

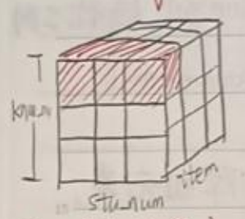
未作答的题，作答结果为-1

a1_cop =



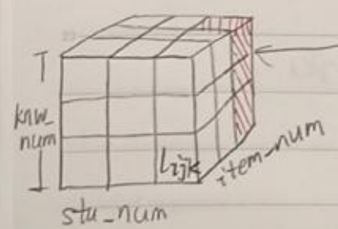
向右旋翻

a1_cop =



P32式(3.24)的$\lambda_{jk}\cdot(1-\alpha_{ik})$中的$\lambda_{jk}$，即"a1_cop矩阵的item_num×knw_num"

tem_yita =



max(temp_yita,[],3)

返回每一深度上的最大值，即学习者i作答试题j时，知识弱项值最大的知识点的知识弱项。用1-它，得P32式(3.22)的δ_{ij}（i对j的掌握程度）

R=r_rand <= r;

满足r_rand矩阵<=r矩阵的位置，R对应位置为1，不满足为0，用来生成R作答矩阵(i×j)。

当学生的真实作答正确的概率>=(随机生成的)题目要求的概率，R为1

反应阶段建模 P31

$$r_{ij} = h(\alpha_i, \Omega_j)$$

@刘 P33 $\Omega_j = [s_j, \lambda_j, g_j]$

输入参数alpha由认知阶段建模得到，大小为stu_num行knw_num列。

a1是@刘畅P35的参数a_j^*，服从Beta分布 @刘畅P36

$1-a_j \sim Beta(\mu_a, S_a)$ 其中$\mu_a=0.5$，$S_a=5.25$。

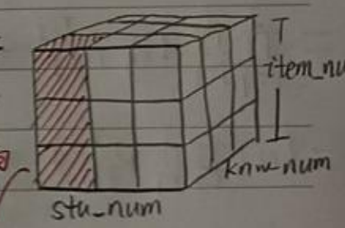
a1参数大小为knw_num行item_num列。a1就是a_j就是$\Omega_j=[s_j,\lambda_j,g_j]$中的$\lambda_j$。

s1，g1参数即s_j，g_j，大小都是1行item_num列。

s1_cop =

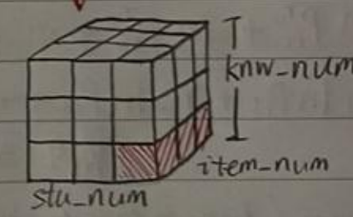
stu_num × item_num
i j

alpha_cop =



向直接翻

alpha_cop = permute(alpha_cop, [1,3,2]) =



P32式(3.21)的$(1-\alpha_{ik})$

即1-"alpha_cop矩阵的stu_num × knw_num"

```
function P=L_CRM1(theta,a1,a2,s1,g1,d211,d212,R,yita,T)
%%choose为函数执行功能选项
%%求theta的似然函数，theta为i的能力参数，yita为i的转换率参数，a1,d1为所有的试题参数，d1,d2为所有的知识点参数。
   d221=d211+(log(0.001)-(log(a2)+log(0.999)))/1.7;  %P36公式
   d222=d212+(log(0.001)-(log(1-a2)+log(0.999)))/1.7;  %P36公式
   alpha=cognitive_m1(a2,d211,d212,d221,d222,theta,yita);
   %alpha_cop=repmat(alpha(i_th,:),item_num,1);
   %d1_cop=repmat(d1,item_num,1);
   [r,~]=response_m1(alpha,a1,s1,g1,T);
   %r=1./(1+exp(-1.7*(alpha(i_th,:)*(a1.*Q')-s1)));
   P=sum(log(r(R==1)))+sum(log(1-r(R==0)));%计算似然值P13,式（2.8）
end
```

L_CRM1.m 计算似然值

P= sum(log(r(R==1))) + sum(log(1-r(R==0))); P13 式(2.8)

$$\ln\left(\prod_{i=1}^{I}\prod_{j=1}^{J} P_{ij}^{r_{ij}} Q_{ij}^{1-r_{ij}}\right)$$

$$= \ln P_{1j}^{r_{1j}} Q_{1j}^{1-r_{1j}} + \ln P_{2j}^{r_{2j}} Q_{2j}^{1-r_{2j}} + \cdots + \ln P_{Ij}^{r_{Ij}} Q_{Ij}^{1-r_{Ij}}$$

$$= \prod_{i=1}^{I} \ln \prod_{j=1}^{J} P_{ij}^{r_{ij}} Q_{ij}^{1-r_{ij}}$$

同理 $\prod_{i=1}^{I}\prod_{j=1}^{J} \ln P_{ij}^{r_{ij}} Q_{ij}^{1-r_{ij}}$

$$= \prod_{i=1}^{I}\prod_{j=1}^{J} \left(\ln P_{ij}^{r_{ij}} + \ln Q_{ij}^{1-r_{ij}}\right)$$

$$= \prod_{i=1}^{I}\prod_{j=1}^{J} \ln P_{ij}^{r_{ij}} + \prod_{i=1}^{I}\prod_{j=1}^{J} \ln Q_{ij}^{1-r_{ij}}$$

$$= \ln\left(\prod_{i=1}^{I}\prod_{j=1}^{J} P_{ij}^{r_{ij}}\right) + \ln\left(\prod_{i=1}^{I}\prod_{j=1}^{J} Q_{ij}^{1-r_{ij}}\right)$$

$$= \left(\ln P_{11}^{r_{11}} + \ln P_{12}^{r_{12}} + \cdots + \ln P_{IJ}^{r_{IJ}}\right) + \left(\ln Q_{11}^{1-r_{11}} + \ln Q_{12}^{1-r_{12}} + \cdots + \ln Q_{IJ}^{1-r_{IJ}}\right)$$

$$= \left(r_{11}\ln P_{11} + r_{12}\ln P_{12} + \cdots + r_{IJ}\ln P_{IJ}\right) + \left(1-r_{11}\right)\ln Q_{11} + \left(1-r_{12}\right)\ln Q_{12} + \left(1-r_{IJ}\right) Q_{IJ}$$

↓ r_{ij}为1的项保留，做对的题保留　　↓ r_{ij}为0的项保留，错的题

= sum(ln对的题的做对概率) + sum(ln错的题的做错概率)

Matlab sum(log(r(R==1))) + sum(log(1-r(R==0)))

∴ L_CRM1.m 文件中函数的返回值是 P13式(2.8) 求 ln

$$P = \ln\left(\prod_{i=1}^{I}\prod_{j=1}^{J} P_{ij}^{r_{ij}} Q_{ij}^{1-r_{ij}}\right) = \ln\left(L(\theta,a,b|R)\right) = \ln L$$

CR_model.m

```
%%%
```

```
%%%
```